

Conditional Width-One Bounds for Generalized Davenport Constants of C_5^3

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Abstract

Let $D_k(G)$ be the least length that forces k pairwise disjoint nonempty zero-sum subsequences. Conditional on the exact inputs $D_3(C_5^3) = 25$, $s_{\leq 6}(C_5^3) = 24$, $s_{\leq 5}(C_5^3) = 33$, and the lower line $D_k(C_5^3) \geq 5k + 10$, we derive $5k + 10 \leq D_k(C_5^3) \leq 5k + 11$ for every $k \geq 4$.

If $D_4(C_5^3) = 30$, then the lower line is exact for every $k \geq 2$. To analyze the remaining alternative, we study a hypothetical total-zero sequence of length 31 with no nonempty zero-sum subsequence of length at most five. For saturated short-free sequences over elementary abelian groups of odd prime exponent, we prove a defect certificate that excludes multiplicity $p - 2$. In the case $p = 5$, a saturated obstruction (in particular, every obstruction with support at least nine) has only multiplicities 1, 2, 4. If s is support size and c_i counts multiplicity i , then $c_2 = 31 - s - 3c_4$ and $c_1 = 2s - 31 + 2c_4$.

The subsequence of points of multiplicity at least two has length $62 - 2s - 2c_4$. Using the short-zero-sum threshold $\eta(C_5^2) = 13$, we prove that this stratum must span rank three whenever $s + c_4 \leq 24$. We also prove an exact identity coupling support deficit, repetitions inside zero-sum atoms, and overlaps between atom supports. Two separate exact implementations found no obstruction through support ten within the supplied computational model. Because this computation has not been independently replicated, we report it only as bounded computational evidence.

The exact value remains $D_4(C_5^3) \in \{30, 31\}$. A larger search through support 22 is also bounded evidence and has no theorem authority.

Keywords: generalized Davenport constants; zero-sum sequences; elementary abelian groups; short zero sums; inverse zero-sum problems

1. Introduction

Generalized Davenport constants measure the sequence length required to force several pairwise disjoint zero sums. Rank-two groups exhibit strong eventual structure, whereas rank-three groups can depart from the expected arithmetic progression.

Under the declared inputs below, the values $D_2 = 20$ and $D_3 = 25$, together with short-zero-sum thresholds, reduce the next value to

$$D_4(C_5^3) \in \{30, 31\}.$$

This one-bit ambiguity controls the tail: the lower candidate $D_4 = 30$ forces $D_k = 5k + 10$ for every $k \geq 2$. The upper candidate would yield a rigid length-31 total-zero sequence with no zero sum of length at most five.

We combine the recurrence with symbolic analysis of that hypothetical obstruction. The purpose is twofold: to obtain a conditional width-one theorem for the full tail and to identify exactly where existing rank reductions cease to be automatic. Computation is kept in a separate evidence class.

Our contributions are:

1. the conditional width-one corridor for every $k \geq 4$;
2. the conditional exact tail if $D_4 = 30$;
3. an odd-prime saturation-defect lemma excluding multiplicity $p - 2$;
4. bounded computational evidence through support ten;
5. an exact multiplicity parameterization for every remaining support stratum;
6. the rank-forcing phase $s + c_4 \leq 24$ for the high-multiplicity stratum; and
7. an identity coupling support deficit to internal atom repetition and cross-atom overlap.

2. Definitions and declared inputs

For a finite abelian group G , $D_k(G)$ is the least integer n such that every sequence of length at least n over G contains k pairwise disjoint nonempty zero-sum subsequences. Let $s_{\leq \ell}(G)$ denote the least length forcing a nonempty zero-sum subsequence of length at most ℓ . For an odd prime p , a sequence is *p-short-free* if it has no nonempty zero-sum subsequence of length at most p . It is *saturated* relative to this property if appending any group element destroys it. Thus *5-short-free* and *saturated* below are the specializations at $p = 5$.

We use the generalized-Davenport recurrence framework [1]

$$D_{k+1}(G) \leq \max\{D_k(G) + \ell, s_{\leq \ell}(G) - 1\},$$

and take the lower bound $D_k(C_5^3) \geq 5k + 10$ in the range considered here as an explicit input. The other exact inputs are

$$D_2 = 20, \quad D_3 = 25, \quad s_{\leq 6} = 24,$$

while $s_{\leq 5}(C_5^3) = \eta(C_5^3) = 33$. We also condition the structural analysis on the inverse-theorem input that every length-32 5-short-free sequence has exactly eight support points. These rank-three statements are explicit hypotheses, not results proved in this manuscript; every corridor statement below is conditional on them. We also use the published rank-two value $\eta(C_5^2) = 13$.

3. A width-one tail

Taking $\ell = 6$ gives

$$D_4 \leq \max\{25 + 6, 24 - 1\} = 31.$$

Together with the lower bound, this proves $D_4 \in \{30, 31\}$.

Theorem 1 (conditional corridor). Under the declared exact inputs, for every $k \geq 4$,

$$5k + 10 \leq D_k(C_5^3) \leq 5k + 11.$$

Proof. The base case is $D_4 \leq 31$. Apply the recurrence with $\ell = 5$ and $s_{\leq 5} - 1 = 32$. If $D_k \leq 5k + 11$, then both terms in the maximum are at most $5(k + 1) + 11$. The declared lower-bound input supplies the lower line. $\square$

Theorem 2 (conditional tail). Under the declared exact inputs, if $D_4(C_5^3) = 30$, then

$$D_k(C_5^3) = 5k + 10 \quad (k \geq 2).$$

Proof. The statement holds for $k = 2, 3, 4$. Applying the same recurrence with $\ell = 5$ propagates the upper bound $5(k + 1) + 10$, which matches the declared lower-bound input. $\square$

There is no converse here: the assumption $D_4 = 31$ does not by itself prove that later terms lie on the upper line.

4. The hypothetical length-31 obstruction

Assume $D_4(C_5^3) = 31$. The extremal characterization of Freeze and Schmid provides a total-zero sequence S of length 31 whose maximum factorization length is four. If M is the minimum length of an atom dividing S , their recurrence gives $31 = D_4 \leq D_3 + M = 25 + M$, so $M \geq 6$. On the other hand, $s_{\leq 6}(C_5^3) = 24$ forces every 31-term sequence to contain a zero-sum subsequence of length at most six; an inclusion-minimal such subsequence is an atom. Hence $M = 6$, and S contains no zero-sum subsequence of length at most five. Thus the upper candidate yields a sequence satisfying

$$|S| = 31, \quad \sigma(S) = 0,$$

with no nonempty zero-sum subsequence of length at most five. Every element has multiplicity at most four. If its support exceeds eight, it is saturated: otherwise a short-free one-term extension would have length 32, whereas the declared inverse-theorem input forces every length-32 5-short-free sequence to have exactly eight support points.

Theorem 3 (saturation defect). Let p be odd and let S be a saturated p -short-free sequence over an elementary abelian exponent- p group. If a nonzero element x has multiplicity $m < p$, then there is a subsequence R such that

$$|R| \leq p - 1 - m, \quad x \notin \text{supp}(R), \quad \sigma(R) = -(m + 1)x.$$

Proof. Append one further copy of x . Saturation supplies a zero-sum subsequence of length at most p that uses the appended occurrence. It must also use every original copy of x ; otherwise replacing the appended copy by an unused original occurrence would give a short zero sum already contained in S . Removing the $m + 1$ copies of x leaves the required R . $\square$

Corollary 4. Multiplicity $p - 2$ is impossible.

Indeed, the defect subsequence would have length at most one and sum x , forcing a forbidden additional copy of x . For $p = 5$, multiplicity three is absent. Singleton and double points also acquire defect certificates of length at most three and two, respectively.

5. Exact multiplicity grammar

For the saturated case $s \geq 9$, Corollary 4 excludes multiplicity three. Let $s = |\text{supp}(S)|$, and let c_1, c_2, c_4 count points of multiplicity one, two, and four. Then

$$c_1 + c_2 + c_4 = s, \quad c_1 + 2c_2 + 4c_4 = 31.$$

Writing $c_4 = j$ gives

$$c_2 = 31 - s - 3j, \quad c_1 = 2s - 31 + 2j.$$

Thus c_1 is odd, and nonnegativity determines the complete admissible range for every support size. For example, the support-23 patterns are

$$1^{15}2^8, \quad 1^{17}2^54, \quad 1^{19}2^24^2,$$

and the support-24 patterns are

$$1^{17}2^7, \quad 1^{19}2^44, \quad 1^{21}24^2.$$

6. The rank-forcing phase

Let H be the subsequence consisting of all points of multiplicity two or four. Its length is

$$|H| = 2c_2 + 4c_4 = 62 - 2s - 2c_4.$$

If H were contained in a subgroup of rank at most two, it would be a 5-short-free sequence over C_5^2 . The value $\eta(C_5^2) = 13$ forces a short zero sum once $|H| \geq 13$. Since $|H|$ is even, the contradiction begins at $|H| \geq 14$.

Theorem 5 (rank-forcing phase). For a saturated obstruction with $s \geq 9$, the high-multiplicity stratum spans rank three whenever

$$s + c_4 \leq 24.$$

Proof. The condition $|H| \geq 14$ is equivalent to $62 - 2s - 2c_4 \geq 14$, hence to $s + c_4 \leq 24$. $\square$

The theorem gives the following exact boundary:

Support	c_4	Consequence from Theorem 5
23	0 or 1	rank three forced
23	2	threshold is silent
24	0	rank three forced
24	1 or 2	threshold is silent
at least 25	any admissible value	threshold is silent

Silence is not a converse. Outside the phase, rank three may follow from a different argument.

7. Bounded low-support exclusion

Symbolic reductions combined with two exact state representations exclude every support size through ten.

Computational result. In the supplied exact search model, no length-31 total-zero 5-short-free sequence over C_5^3 exists with support at most ten.

Exact search description. Multiplicity at most four first excludes supports at most seven. At support eight, the only length pattern is 4^73 . Adding the missing fourth copy of the triple point preserves short-freeness; total sum then forces that point to equal the negative support sum. After normalizing a basis by $\text{GL}(3, 5)$, two exact subset-sum representations enumerate all 564 normalized supports and find none satisfying this condition.

For supports nine and ten, saturation and Corollary 4 exclude multiplicity three. The support-nine equations leave only 4^721 ; a canonical search with the last point forced by total sum visits 6,537,270 states in each of two independent exact-weight subset-sum engines and finds no solution. At support ten, the only patterns are 1^34^7 and $1, 2^34^6$. Four multiplicity-four points must span rank three because 16 terms in a rank-two subgroup would contradict $\eta(C_5^2) = 13$. Normalizing an independent triple to the standard basis makes the remaining enumeration complete. The two engines agree exactly: they visit respectively 210,700 states with 3,558 leaves and 272,119 states with no leaves, finding zero solutions in both patterns.

Both engines prune a partial candidate exactly when adding a term creates a zero-sum subsequence of one of the lengths one through five; one stores explicit byte reachability by weight and group sum, while the other stores translation masks in a 128-bit representation. Their source, build instructions, expected rows, and machine-readable results are included as ancillary files. Thus every support stratum through ten is exhausted within the declared model. Because a clean-room external replay has not been completed, this result is not used as theorem authority.

A larger bounded computation found no such sequence through support 22. That search has not received an external independent replay and is not used in any proof above or below. Its bounded conclusion is useful for prioritizing the next case split, but it does not establish support at least 23 as a theorem.

8. Atom repetition and overlap

Conditionally suppose that an independently verified extremal-factorization argument establishes one of the candidate atom-length types

$$(6, 6, 6, 13) \quad \text{or} \quad (6, 6, 7, 12).$$

The identity below then applies; in fact, it holds for every occurrence-disjoint atom factorization.

Let $S = U_1 \cdots U_r$ be an occurrence-disjoint atom factorization. For each group element g , let r_g be the number of atom supports containing g , and define the internal deficit

$$\delta_i = |U_i| - |\text{supp}(U_i)|.$$

Theorem 7 (repetition-overlap identity). Every occurrence-disjoint factorization satisfies

$$|S| - |\text{supp}(S)| = \sum_i \delta_i + \sum_{g \in \text{supp}(S)} (r_g - 1).$$

Proof. Summing the atom support sizes counts every global support point r_g times. Hence

$$\sum_i \delta_i = |S| - \sum_g r_g,$$

while cross-atom overlap equals

$$\sum_g r_g - |\text{supp}(S)|.$$

Adding the two identities proves the claim. $\square$

Thus any independently established lower support bound s_0 gives a total internal-repetition plus cross-overlap budget of at most $31 - s_0$. The internally reproduced support-ten computation suggests $s_0 = 11$ and hence budget 20, but that numerical consequence remains computational evidence. If the bounded support-22 computation is later independently certified, the budget sharpens to eight; that sharper value is conditional and not used as theorem authority here.

9. The residual obstruction phase

A decisive next analysis should impose simultaneously:

1. the multiplicity pattern from Section 5;
2. the rank status from Theorem 5;
3. the saturation-defect certificate at every singleton and double point;
4. one of the two conditional atom-length types;
5. the repetition-overlap identity; and
6. total sum zero with exclusion of zero sums of lengths one through five.

This intersection is substantially smaller than an unstructured support search. A surviving sequence would require independent factorization verification before establishing $D_4 = 31$. A complete, independently replayed infeasibility proof for the length-31 target would establish the lower candidate $D_4 = 30$.

10. Relation to prior work

The generalized-Davenport recurrence, lower-bound framework, extremal factorization language, and eventual-linearity context are established results [1]. The C_0 framework and short-zero-sum localization are also prior work [2], as are the inverse zero-sum program [3] and rank-two inverse results [4].

The residual contribution is the width-one synthesis for C_5^3 , the saturation-defect specialization, the exact multiplicity grammar, the rank-forcing phase, and the coupling of support deficit with atom repetition and overlap. These are structural compression results; they do not decide the final extremal bit.

11. Reproducibility and limitations

The conditional corridor, defect lemma, and support-ten exclusion are separately bound to exact records. A separate implementation enumerates all multiplicity patterns for supports 8-31, checks the rank-phase equivalence, reproduces the support-23 and support-24 tables, and validates the atom identity on independently generated factorizations. Only the displayed analytic proofs carry all-parameter authority.

The exact value of $D_4(C_5^3)$ remains open, as does the associated length-31 extremal-spectrum statement. The support-22 frontier is bounded computation pending external replay. The rank phase is one-way, and the atom identity is a compression principle rather than a contradiction. No additional implementation can substitute for an independently auditable final obstruction proof.

12. Conclusion

Conditional on the declared exact inputs, the generalized Davenport constants of C_5^3 remain within one unit of the line $5k + 10$ for every $k \geq 4$, and the lower choice at $k = 4$ would fix the entire tail. The hypothetical upper-line obstruction has substantially more structure than this corridor alone reveals: its multiplicities have an exact grammar, its high-multiplicity stratum enters a sharp rank-forcing phase, and its atom factorization obeys a global repetition-overlap budget.

These results explain where the residual difficulty begins and define a smaller intersection problem for the unresolved extremal bit. They do not claim to have resolved it.

Data and code availability

The verification package contains exact multiplicity enumeration, rank-phase checks, atom-identity tests, and separately labeled bounded-search records. These files are distributed in the source archive accompanying this version.

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